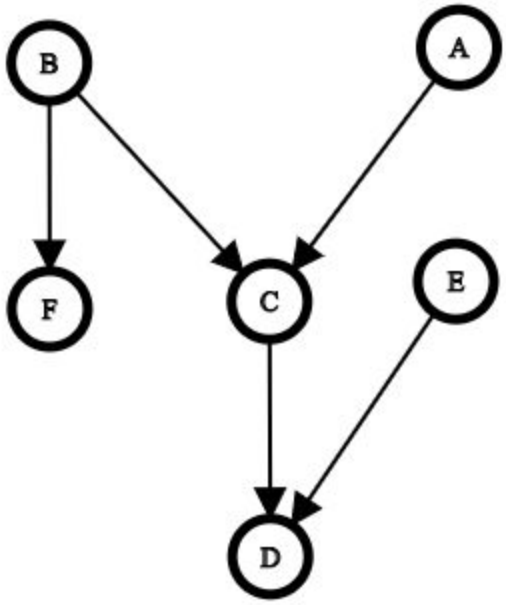
1. Consider the following Bayesian Network. Suppose we wish to compute P(C|A,B,D,F,E) for running MCMC with Gibbs Sampling.



Which of the following represents the simplified problem that needs to be solved? In all of the following options, ***α*** is the normalizing constant.

a. P(C|A,B,D,F,E) = ***α***. P(C|A). P(C|B)
b. P(C|A,B,D,F,E) = ***α***. P(C|A,B)
c. P(C|A,B,D,F,E) = *α*. P(C|A,B). P(D|C,E)
d. P(C|A,B,D,F,E) = ***α***. P(C|A,B). P(C|D)

We know, a random variable is independent of all other variables given its Markov Blanket. Therefore, P(C|A,B,D,F,E) = P(C|A,B,D,E) as {A,B,D,E} is Markov Blanket of C.

Now, P(C,A,B,D,E) = P(C|A,B).P(D|C,E)

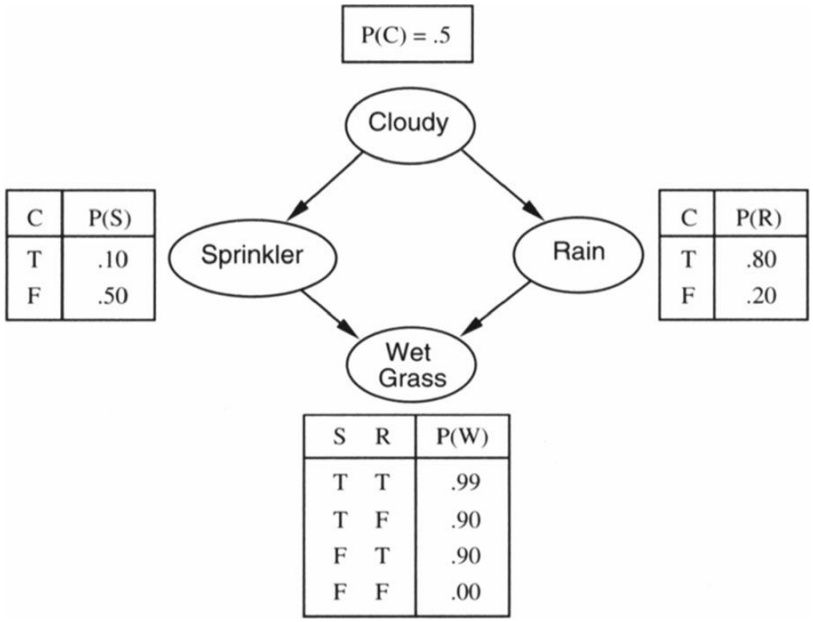
P(C|A,B,D,E) = P(C,A,B,D,E)/ P(A,B,D,E)

P(A,B,D,E) = P(A).P(B). {P(D|E,C) + P(D|E,-C)} which is independent of C.
Therefore, P(C|A,B,D,E) = ***α***. P(C,A,B, D, E) = ***α***. P(C|A,B). P(D|C,E)

2. Which of the following sampling algorithms will ensure that no samples are discarded?
    a. Rejection Sampling
    b. Likelihood Sampling
    c. Gibbs Sampling

    Likelihood Sampling and Gibbs Sampling don't reject any samples since samples are generated in such a way that all the observed variables are set to their given values explicitly. Refer class notes for details and examples.

3. Consider the following Bayesian Network. Suppose you are doing likelihood sampling to determine P(S|¬C,W).

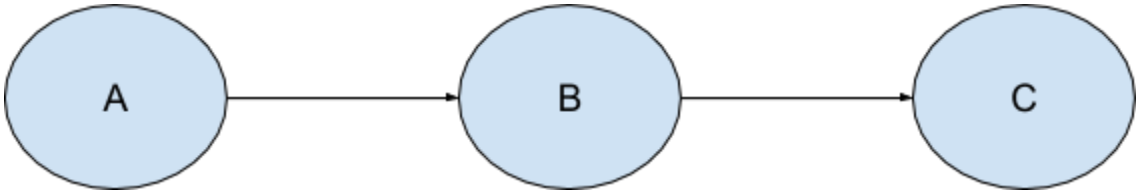


    Let the weight for the sample (¬C,S,¬R,W) be w. **What is 100w?** (Round off your answer to the closest integer)
    Ans. 45

    Weight for a sample <z,e> = $\prod$ P(e_i| Parents(e_i))

    Weight of sample (¬C,S,¬R,W) = P(¬C).P(W|S,¬R) = 0.5*0.9 = 0.45

4. Consider the following Bayesian Network.



We are given the following samples.

A	B	C
1	0	1
0	1	1
0	1	1
1	0	1
0	0	1

What is the estimated value of P(C=1|B=0) if we use Add-One smoothing? Round off your answer till 2 decimal points. (e.g. if your answer is 0.14159… then write 0.14)

Ans. 0.80

P(C=1|B=0) = P(C=1,B=0)/ (P(C=1,B=0)+P(C=0,B=0)) Bayes' Rule

= 3/(3+0) = 1

But we are using add-one smoothing. So instead we need to do

P(C=1|B=0) = P(C=1,B=0)/ (P(C=1,B=0)+P(C=0,B=0)) Bayes' Rule

= (3+1)/{(3+1)+(0+1)} = ⅘

Error in answer on NPTEL portal. Question to be deleted.

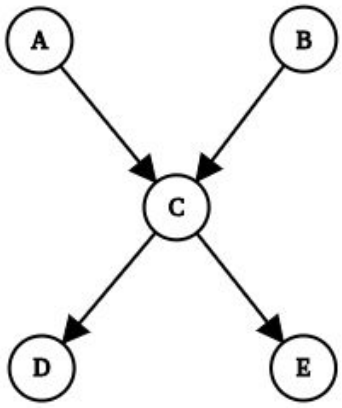
5. Suppose you toss a coin 100 times and find that head occurs 72 times. You use maximum likelihood estimate to obtain P(H) to be p. However, you also know that the typical P(H) is close to 0.5 and you use this prior knowledge to do maximum a posteriori estimate. The estimate of P(H) obtained using MAP estimate is q. What is the relationship between p and q?
   a. **p>q**
   b. p<q
   c. p=q
   d. Not enough information

For Maximum Likelihood, P(H) = 72/100 = 0.72

For Maximum A Posteriori, P(H) = $\text{argmax}_q \{P(D|q).P(q)\}$

P(q) will peak at 0.5 while P(D|q) will peak at 0.72 as a result P(H) will attain maxima somewhere in between 0.5 and 0.72. As a result $P(H)_{ML} > P(H)_{MAP}$

6. Consider the following Bayesian Network where each variable is binary.



Consider the following training examples for the above bayesian net where two examples contain unobserved values i.e. D_7 and E_8. Initialize the 10 parameters of the network at 0.5. Find $P(D_7 = 1) * P(E_8 = 1)$ after simulating the first E step of the simple EM algorithm. (upto 2 decimal places)

#	A	B	C	D	E
1	1	0	1	1	1
2	0	1	1	1	0
3	1	1	1	1	1
4	0	0	0	0	0
5	0	0	0	1	0
6	0	0	0	0	1
7	1	1	1	?	1
8	1	1	1	1	?

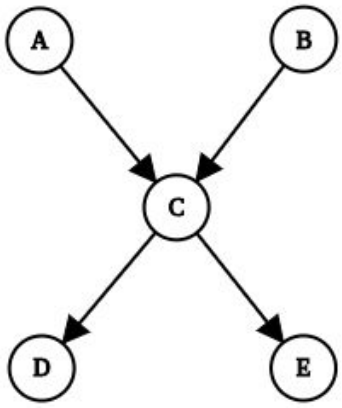
Ans: 0.25

$P(D_7=1) = P(D_7=1 \mid A_7=1, B_7=1, C_7=1, E_7=1) = P(D_7=1 \mid C_7=1) = P(d \mid c) = 0.5.$

$P(E_7=1) = P(E_7=1 \mid A_7=1, B_7=1, C_7=1, D_7=1) = P(E_7=1 \mid C_7=1) = P(e \mid c) = 0.5.$

Both P(d | c) and P(e | c) are 0.5 as they are 1 of the 10 parameters of the network and are initialized to 0.5. $P(D_7 = 1) * P(E_8 = 1) = 0.25$.

7. Consider the following Bayesian Network, where each variable is binary.



Consider the following training examples for the above Bayesian net where two examples contain unobserved values i.e. D_7 and E_8. Initialize the 10 parameters of the network at 0.5. Find $P(D_7 = 1)$ after simulating the second E step of the simple EM algorithm. (up to 1 decimal places)

#	A	B	C	D	E
1	1	0	1	1	1
2	0	1	1	1	0
3	1	1	1	1	1
4	0	0	0	0	0
5	0	0	0	1	0
6	0	0	0	0	1
7	1	1	1	?	1
8	1	1	1	1	?

Ans: 0.9

Using the previous question, D_7 can be 1 with probability 0.5 or basically, it can be replaced by 0.5. Now using the M step, we have

P(d | c) = P(D = 1 | C = 1) = n(D = 1, C = 1) / n(C = 1) = (1 + 1 + 1 + 0.5 + 1) / 5 = 0.9
Using the E step similarily as in previous question, we have,
$P(D_7=1) = P(d \mid c) = 0.9$.

8. Consider the following samples of two variables A and B. Here A is binary and B is a four-valued variable, and # represents the number of times this state was seen in the data.

A	B	#
0	1	4
1	1	1
0	2	0
1	2	5
0	3	6
1	3	2
0	4	0
1	4	1

In general, we can do add-alpha smoothing. For which of the following values of alpha, P(B = 2 | A = 1) is at least 0.5.

   a. 0.15
   b. 0.30
   c. 0.45
   d. 0.60

$$P(B=2 \mid A=1) = \frac{P(B=2, A=1)}{P(A=1)} = \frac{n(B=2, A=1}{n(A=1)} = \frac{5+}{1+ \ +5+ \ +2+ \ +1+} > 0.5$$

$$\frac{5+}{9+4} > 0.5$$

$$< 0.5$$

Therefore, options A, B, C are correct.

9. Which of the following statement(s) is/are correct for ML and MAP estimation learning?
   a. MAP estimate can be used if no prior knowledge about the parameters is known.
   b. MAP and ML are the same if P(θ) is a constant independent of θ.
   c. MAP and ML are the same in the limit of infinite training data.
   d. Both MAP and ML output a single best set of parameters.

ML: $\text{argmax}_{\Theta} P(D \mid \Theta)$
MAP: $\text{argmax}_{\Theta} P(D \mid \Theta).P(\Theta)$
Option A is incorrect since MAP requires prior knowledge about the parameters. Option B is correct because if $P(\Theta)$ is constant then both formulae become the same. Option C is correct because, in the limit of infinite training data, the $P(D \mid \Theta)$ term overweighs the prior and results in both the estimate converging to the same values. Well, it is not that clear yet intuitive. It is possible to derive the result but we prefer to skip it here. Note that the prior probability should be positive in MAP for each case otherwise the statement id obviously true. Though it has been skipped here, it is assumed explicitly. Option D is trivially true from the formula.

10. Consider an intelligent interactive language tutoring system. Which of the following properties of an AI environment does it have?
    a. Deterministic
    b. Sequential
    c. Dynamic
    d. Fully Observable

The environment is not deterministic because, for a given state and action, the next state (that could possibly include some input from the user) cannot be determined accurately. Since it is a tutoring system, it is expected to remember a past couple of learning sessions rather than just the current one to teach. Hence, it is sequential, not episodic. The environment is dynamic as the responses made by the user are dependent on a lot of factors like mood, stress, ease of concept being taught, etc. The environment is partially observable as the system would need to maintain a belief of the user's understanding of different concepts and act accordingly.